TRUEGENE

Legal Safety Platform for Synthetic Biology

Ensuring Ethical & Legally Safe Synthetic Biology

VENTURE OVERVIEW

TrueGene is an SaaS platform that bridges the critical gap between biotechnology research and international legal safety. By providing automated risk assessment tools, we are enabling researchers to identify potential compliance and patent overlaps during the early stages of development, long before significant capital is committed, through an innovative platform.

We are Team TrueGene, a 5-member student founding team out of PES University's Centre for Innovation & Entrepreneurship. Our working prototype is live at bio-sphere-origin.vercel.app

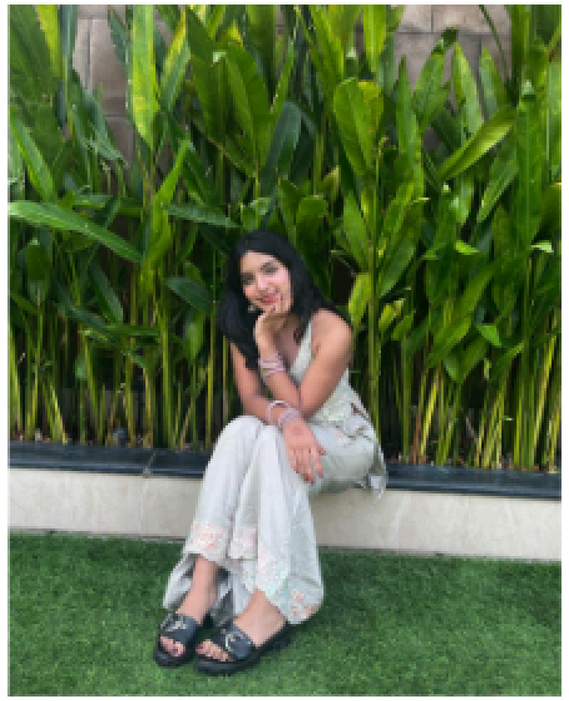
THE TEAM

Shreya Kandrika

I’m Shreya, a biotechnology engineering student with a versatile skillset, and co-founder of TrueGene. Interested in scientific research, law, and bio-entrepreneurship, the crucial intersection between these fields led me to start this venture.

Avantika Kishore

I’m Avantika, a Biotechnology student with a multidisciplinary approach, creative & product strategist, and co-founder of TrueGene. I focus on building the product and positioning it around DNA compliance and safety.



Aditi Krishnan

I’m Aditi, a Biotechnology student and co-founder of TrueGene, with a strong interest in genetics and molecular biology. I focus on bringing that perspective into what we’re building, and I hope to pursue research in the future, which drives my work here.

Maanya Vinayak

I’m Maanya, a Biotechnology student and co-builder of TrueGene. I work on research and shaping our solution around genetic data traceability.

Arjun Subbaraman

I’m Arjun, a computational biologist and the co-founder of TrueGene. While engineering molecular software, I uncovered a massive legal bottleneck in biotech. Today, I’m building the engine to solve it.

DID YOU KNOW?

Biotech innovation is accelerating, but legal compliance is not keeping pace, exposing researchers to significant financial and regulatory risk.

The case that made us ask: what if researchers had known before they built?

The Ozempic / Semaglutide Patent Crisis:

The Incident

- Patent Trap: While the primary patent expired on March 20th 2026, Novo Nordisk's monopoly remains protected by active secondary patents.
- Strict Penalties: Clearance failure triggered aggressive enforcement, immediate sales bans, and total profit forfeiture.

The Impact

- Legal: 100% profit forfeiture.
- R&D: Total capital write-offs.
- Market: Forced product recalls.

₹21 Crore / Month

Lost Profit due to Patent Infringement

₹450 Cr

Total R&D Value at Risk

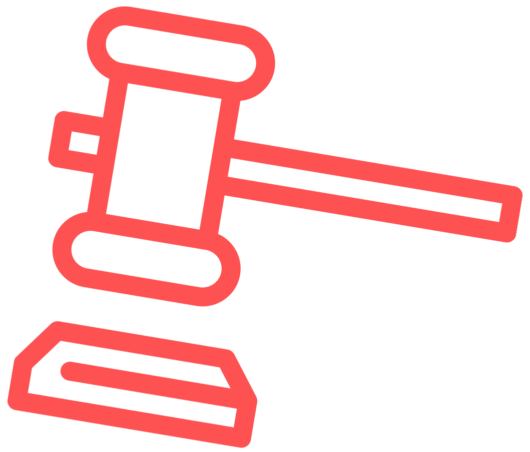
₹1500 Cr

Annual Economic Loss

We let the researchers do the research and not worry about the legal aspect of their research

DNA is designed first.
Compliance is checked later.

By then, it's already too late.



Legal risk is discovered after time, money, and research have already been invested.

CURRENT PROBLEMS

THE WORKFLOW

Biotech researchers design sequences without knowing if they're legally safe.

- Design genetic sequence
- Manual NCBI + WIPO check
- IP lawyer (weeks later, ₹80k+)
- Conflict found

WHY IT FAILS

Every step is a friction point and a money leak.

- Manual Check: Time-consuming
- IP Legal: Expensive
- Late Discovery: Wasted Capital

THE CONSEQUENCE

By the time conflict is found, it's already too late.

68%
of conflicts found too late to pivot

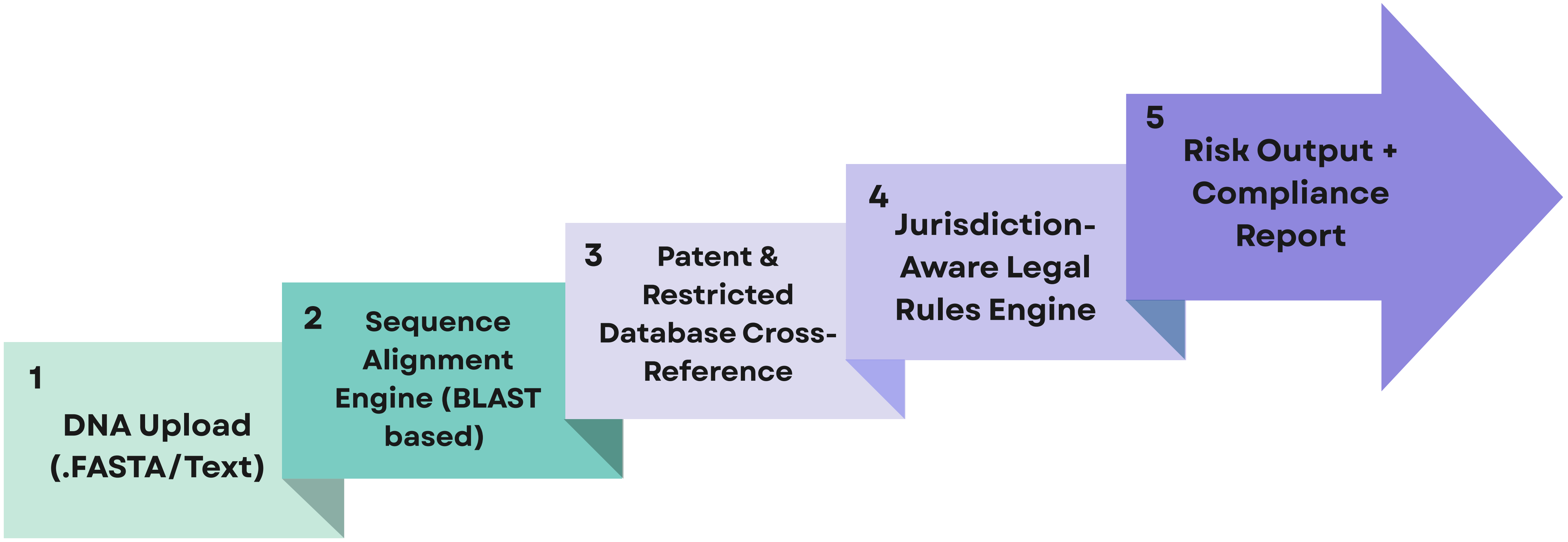
2.3 avg
redesigns per sequence due to IP conflicts

₹11,600 Cr
R&D at risk annually

3+ yrs
avg litigation time

TRUEGENE

A web-based DNA Legal Risk Screening Platform that evaluates genetic sequences for patent conflicts and regulatory compliance before laboratory research begins.



Upload a sequence.
Get a risk score in minutes.

- Green – safe to proceed
- Yellow – licensing review needed
- Red – active patent conflict

Specific results users get

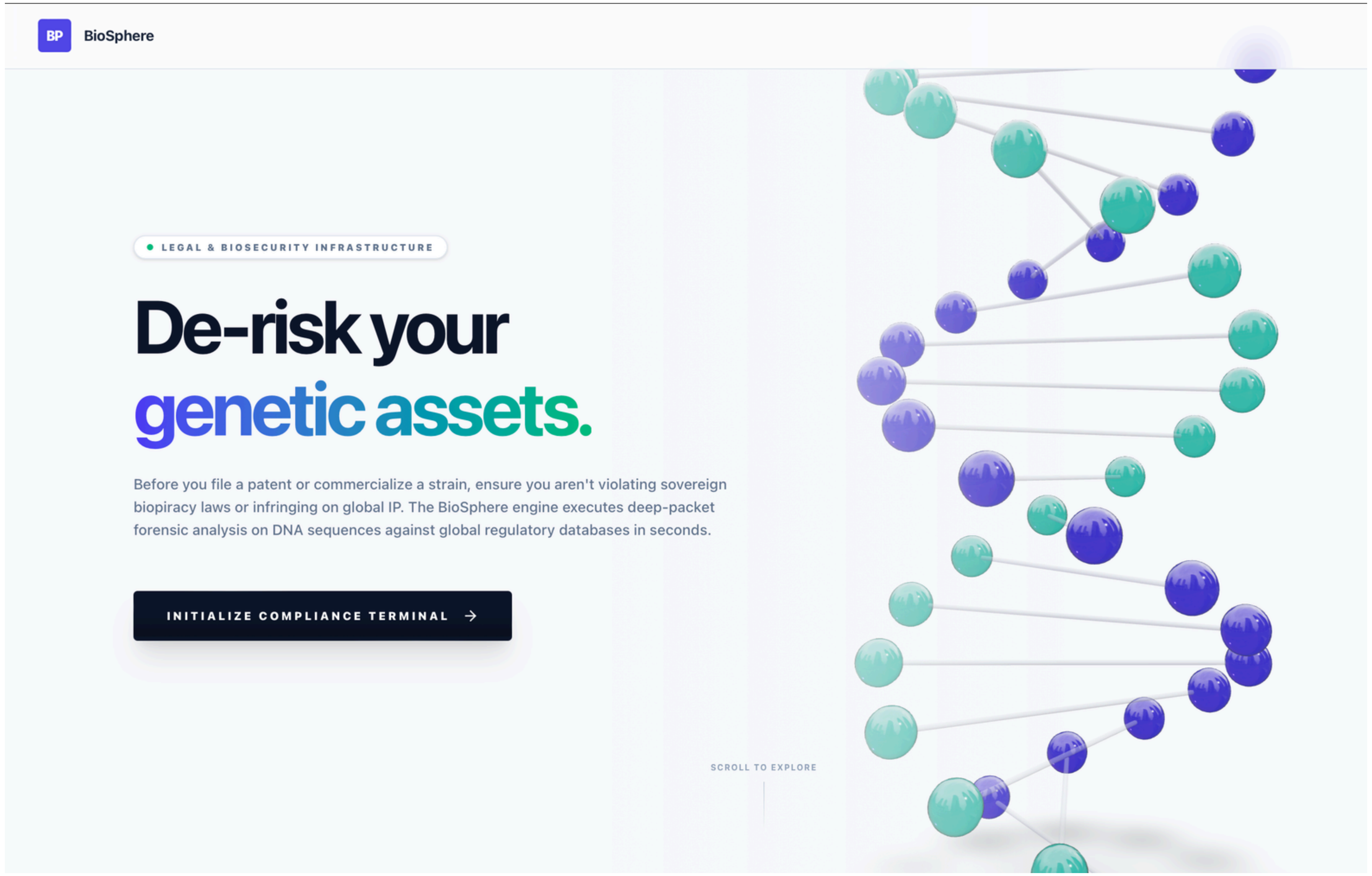
- Immediate Legal Safety Score for any DNA sequence
- Early awareness of licensing or compliance requirements

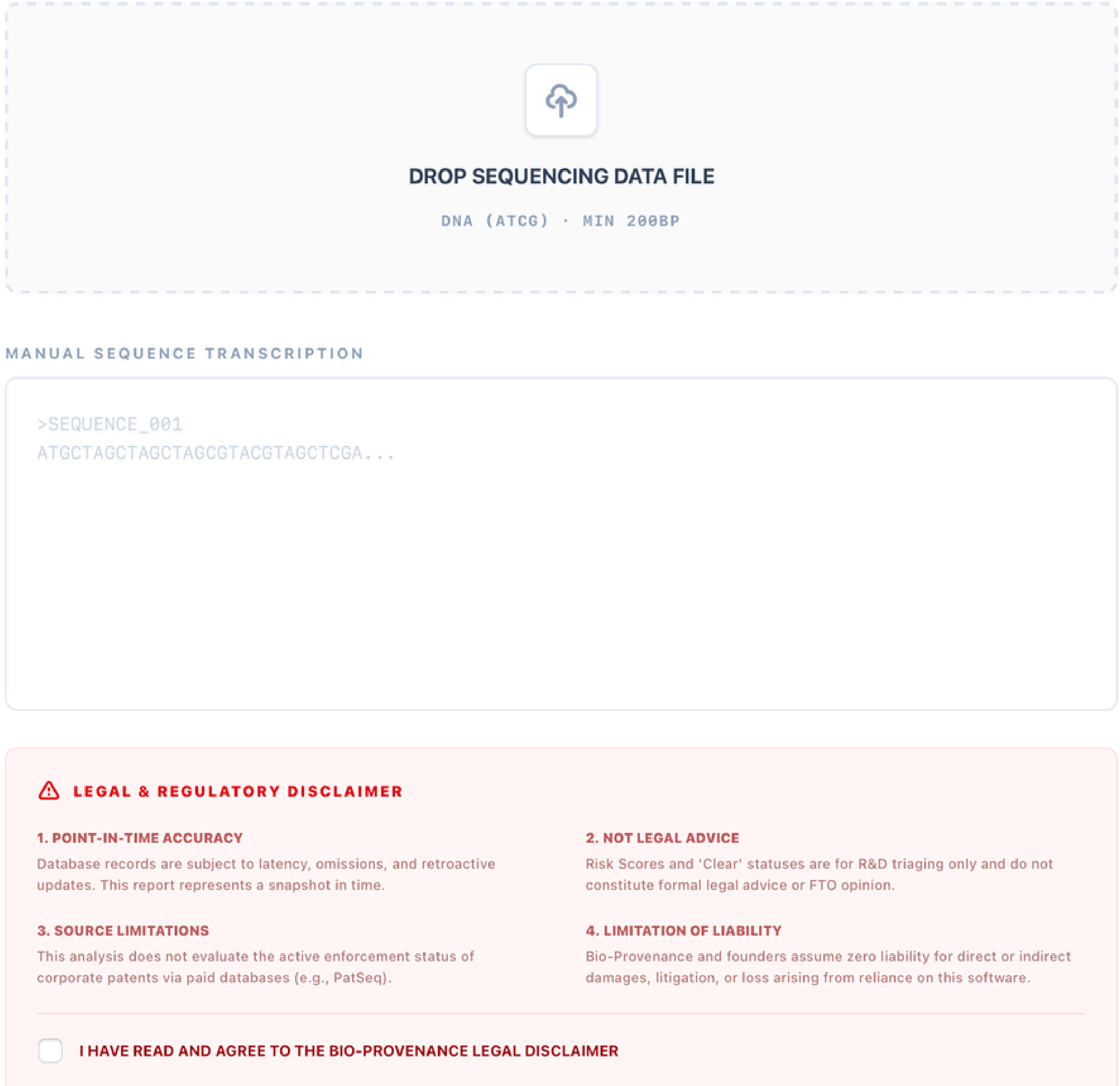
Value proposition for customers

- Prevents expensive patent infringement and biopiracy violations
- Reduces wasted R&D time and financial loss
- Increases confidence for investors and regulatory approvals

A comprehensive report, ready for regulatory submission. TrueGene.

bio-sphere-origin.vercel.app





Our demo video:

https://drive.google.com/file/d/131UyGwtLPkJ_XHPglE3VMQzXV2toCiUn/view?usp=sharing

CUSTOMER SEGMENT

The potential market that we are tapping into:
A rapidly expanding Indian pharma patent market, driven by global patent expiries an biosimilars growth, with an estimated $10 billion opportunity by 2029.

Independent Researchers

01 Instant IP Risk Detection

Research Institutions

02 Ensure Ready, Compliant Research

Biotech & Pharma Companies

03 Enterprise-Grade Legal Protection & Data Security

Built for high-risk, compliance-heavy genetic research workflows.

Backed by Customer Discovery

- Large pharma already has in-house legal teams
- Early-stage biotech & researchers lack affordable IP screening
- Researchers would prefer an early-stage risk indicator before consulting lawyers.
- Strong demand for early-stage risk detection before legal consultation

COMPETITOR LANDSCAPE

Researchers must stitch together multiple tools :
compliance is fragmented, slow, reactive and expensive.

Feature	Public Bio Data (NCBI)	Legal Search (Google / WIPO)	Legacy IP Tools (GenomeQuest)	Bio-Sphere
Access to Sequence Data	✅	❌	✅	✅
Real-Time Compliance Alerts	❌	❌	❌	✅
NBA (India) Specific Checks	❌	❌	❌	✅
Zero-Data Retention (Browser-Local)	❌	❌	❌	✅
Embedded in Lab Workflow	❌	❌	❌	✅
Automated PDF Audit Reports	❌	❌	✅	✅
Automated Bio-Legal Translation	❌	❌	❌	✅

https://www.questel.com/resourcehub/a-guide-to-bio-sequence-patent-searching/

https://www.patsnap.com/resources/blog/articles/master-degenerate-sequence-patent-search-for-ip-teams/

PROGRESS SO FAR

The Spark

PHASE 1

- The Synthetic Biology field as a whole can be considered. Not just a genetic sequence
- Its an infinite field
- Genetics is complex
- Eventually, testing a plug-in with bioinformatics genetics tools

The Build

PHASE 2

- Interviewed **7 domain experts, geneticists, IPR lawyers, patent agents**
- Surveyed **120** biotech students & PhD scholars
- Built the first **working prototype**

The Refine

PHASE 3

- **User tested** the prototype with real researchers
- **Expanded scope from genetics to all of synthetic biology**

The Proof

Won 1st runner-up at PES University's CIE Ignite 2026

- Pitching live to prove TrueGene isn't just viable, **it's necessary**.
- Emerged from a competitive cohort of 500+ teams

https://forms.gle/xmB4VjLfrqvRyXek7

IMPACT AND VISION

A world where no breakthrough is lost to a legal landmine.

THE RESEARCHER
Every researcher gets instant legal clarity before experiments begin – no lawyers, no waiting, no wasted effort.

THE LAB
Labs eliminate costly redesigns and late-stage IP conflicts, saving time and capital on every project they run.

THE INDUSTRY
Fewer drugs and therapies get derailed mid-development – accelerating the pipeline from discovery to patient.

THE NATION
Biopiracy drops. Sovereign genetic resources are safeguarded. India earns its place as a trusted global biotech leader.

THE WORLD
TrueGene embeds legal compliance into every lab workflow worldwide – so no breakthrough is ever lost to a conflict that could have been caught on day one.

WHY US?

Multidisciplinary by design
Biotech, bioinformatics, engineering, and business – we understand the science and can build and scale the product.

Research-backed, not assumption-driven
7 experts across genetics, IP law, and patent practice. 120+ survey responses. We mapped real workflows.

Functional prototype, not a concept deck
A working compliance engine with DNA upload, alignment, risk output, and reporting – built and ready to scale.The foundation is built. We're here to scale it.

WHY NOW?

Synthetic biology is scaling fast
The global synbio market is projected to reach ~$50B by 2030 – dramatically increasing sequence volume, filings, and associated risk.

A growing IP bottleneck
A wave of drug patent expiries (2026–2028) is triggering biosimilar innovation and legal complexity at scale.

India's biotech is accelerating without compliance infrastructure
$130B sector in 2024, projected $300B by 2030. No India-specific DNA compliance layer exists today. We're building it, from Bangalore, the Biotech capital.

As biology becomes programmable, compliance cannot remain manual.
We're building the layer that makes it scalable.

THANK YOU

Because ‘oops’ is expensive in biotech.